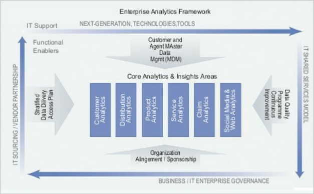

Figure 3: Enterprise analytics planning framework
(Image credits: Google Images)

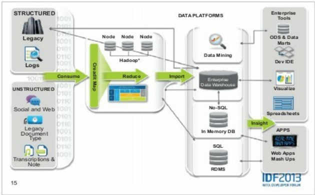

Figure 4: Work flow diagram for the Big Data enterprise model
(Image credits: Google Images)

form, as in the case of transaction information from several online applications or other master data.

Acquire: After various types of data sets are taken from several sources and inserted, they can either be written straight away to real-time memory processes or can be written as messages to disk, database transactions or files. Once they are received, there are various options for the persistence of these data. The data can either be written to several file systems, to RDBMS or even various distributed-clustered systems like NoSQL and Hadoop Distributed File System.

Organise: This is the process of organising various acquired data sets so that they are in the appropriate form to be analysed further. The quality and format of data is changed at this stage by using various techniques to quickly evaluate unstructured data, like running the map-reduce process (Hadoop) in batch or map-reduce process (Spark) in memory. There are other evaluation options available for real-time streaming data as well. These are basically extensive processes which enable an open ingest, data warehouse, data reservoir and analytical model. They extend across all types of data and domains by managing the bi-directional gap between the new and traditional data processing environments. One of their most important features is that they meet the criteria of the four Vs — a large volume and velocity, a variety of data sets, and they also help in finding value wherever our analytics operate. In addition to that, they also provide all sorts of data quality services, which help in maintaining metadata and keeping a track of transformation lineage as well.

Analyse: After the data sets are converted to an organised form, they are further analysed. So the processing output of Big Data, after having been converted from low density data to high density data, is loaded into a foundation data layer. Apart from the foundation data layer, it can also be loaded to various data warehouses, data discovery labs (sets of data stores, processing engines and their analysis tools), data marts or back into the reservoir. As the discovery lab requires fast connections to the event processing, data reservoir and data warehouse, a high speed network like InfiniBand is required for data transport. This is where the *reduction-results* are basically loaded from processing the output of Big Data into the data warehouse for further analysis.

We can see that both the reservoir and the data warehouse offer in-situ analytics, which indicates that analytical processing can take place at the source system without the extra step needed to move the data to some other analytical environment. SQL analytics allows for all sorts of simple and complex analytical queries at each data store, independently. Hence, it is the point where the performance of the system plays a big role as the faster the data is processed or analysed, the quicker is the decision-making process. There are many options like columnar databases, in-memory databases or flash memory, using which performance can be improved by several orders of magnitude.

Decide: This is where the various decision-making processes take place by using several advanced techniques in order to come to a final outcome. This layer consists of several real-time, interactive and data modelling tools. They are able to query, report and model data while leaving the large amount of data in place. These tools include different advanced analytics, in-reservoir and in-database statistical analysis, advanced visualisation, as well as the traditional components such as reports, alerts, dashboards and queries.

Significance and role of Big Data for enterprise applications

Big Data has been really playing quite a significant role in a number of enterprise applications, which is why large enterprises are spending millions on it. Let's have a look at a few scenarios where these enterprises are benefiting by implementing Big Data techniques.

1. The analysis and distillation of Big Data in combination with various traditional enterprise data, leads to the development of a more thorough and insightful understanding of the business, for enterprises. It can lead to greater productivity, greater innovation and a stronger competitive position.
2. Big Data plays a much more important role in healthcare services. It helps in the management of chronic or other